Development of a System for Detecting AI-Generated Images Using Deep Learning Methods

Afanasyev Kirill and Amangeldi Beksultan

Astana IT University, Big Data Analytics, BDA-2403
Astana, Kazakhstan

241064@astanait.edu.kz; 241227@astanait.edu.kz

Abstract

Detectors that recognize images from familiar generators may struggle when the generator or image processing changes. We tested unseen-generator detection and processing robustness with a fine-tuned 50-layer residual network (ResNet-50) and a frozen Contrastive Language-Image Pre-training (CLIP) encoder with a linear classifier. After exact deduplication, the entire third-party Tiny-GenImage mirror contained 34,998 images: 19,998 for training, 4,000 for validation, 6,000 for seen testing and 5,000 for unseen testing. Six generators supplied development data, with Midjourney excluded from training and validation. Both models used three training seeds and identical test IDs. ResNet-50's mean F1 fell from 0.910 on seen generators to 0.310 on Midjourney; CLIP's fell from 0.949 to 0.746. The unseen CLIP-minus-ResNet F1 difference was 0.436, with a paired 95% confidence interval of [0.415, 0.456]. Their unseen false-positive rates were 0.096 and 0.061, respectively. Joint Photographic Experts Group (JPEG) recompression at quality 65 reduced unseen F1 to 0.289 and 0.687. Downsampling to 112×112 raised unseen false-positive rates to 0.971 and 0.561. Neither model met all targets: unseen F1 ≥ 0.80, false-positive rate ≤ 0.10 and seen-to-unseen F1 loss ≤ 0.10. CLIP detected more unseen images and made fewer false alarms on the original inputs, but both models remained sensitive to processing. Unequal fitting budgets, unmatched source characteristics and reuse of previously inspected mirror content limit these findings.

Keywords: AI-generated images; deep learning; image forensics; generator generalization; processing robustness; false-positive rate

1 Introduction

A convincing synthetic image can be difficult to verify because its appearance says little about its origin. The National Institute of Standards and Technology treats detection, provenance tracking and content labelling as complementary approaches to digital-content transparency [1]. Nightingale and Farid found that participants in controlled face-perception experiments struggled to distinguish synthesized faces from real ones [2]. Their study concerns faces, but it illustrates the limits of visual judgement. Automated detectors make mistakes too. A false positive can cast doubt on a genuine photograph, while a missed detection leaves a generated image unflagged. A detector used to assess online content needs to account for both errors.

A low false-positive rate can also be misleading: a model that labels nearly everything real will miss most generated images. Precision, recall and F1 describe this trade-off, and the error counts show how the model reached its score.

Earlier benchmarks show how processing can affect detection. FaceForensics++ tests facial manipulation at different compression levels and image sizes. Corvi et al. examine diffusion-generated images and test compression and resizing [3, 4]. Both include processing in the evaluation, though they study different tasks. Evidence from manipulated faces may not transfer to entirely generated objects or landscapes.

Frank et al. identify spectral patterns that distinguish generated images, and Durall et al. trace spectral discrepancies to upsampling operations [5, 6]. Such patterns may change when an image is resized. These studies give us a reason to test processing, although they do not establish which features cause errors in our classifiers.

Transfer methods use different kinds of evidence. Learning on Gradients and Neighboring Pixel Relationships look for reusable artifacts in network gradients and local pixel dependencies [7, 8]. DIRE and AEROBLADE measure reconstruction error through a diffusion model or an autoencoder [9, 10]. The former methods learn discriminative artifact representations; the latter assess how a generative model reconstructs the input. Each approach makes different assumptions about which traces survive a source change, and their different protocols prevent a common ranking.

Studies using pretrained vision-language features also differ in how much adaptation they recommend. FatFormer argues that a frozen CLIP encoder with a linear head can miss forgery evidence and introduces forgery-aware adaptation [11]. Cozzolino et al. obtain broad transfer with a lightweight CLIP-based detector, including on unfamiliar generators and processed inputs [12]. Their training and evaluation choices differ. Cozzolino et al. also show that strong ranking scores can coexist with weaker accuracy at a fixed threshold, so the operating point needs separate attention.

We examine whether transfer gains are accompanied by acceptable false-positive rates at the chosen threshold, including after processing. Corvi et al. and Cozzolino et al. already test unfamiliar generators and processed images [4, 12]. Our comparison focuses on seed variation and paired changes in F1 and false positives for two configurations evaluated on identical test IDs. Generator changes can coincide with differences in photographic source, category or resolution. We therefore cannot isolate generator identity as the cause of an error.

We compare unseen-generator detection and processing robustness while measuring false positives on real photographs. The study addresses two questions:

1. **RQ1:** How well do ResNet-50 and the CLIP linear classifier detect a generator excluded from development, and what false-positive rates accompany their F1 scores?

2. **RQ2:** How do JPEG compression and downsampling change F1 and false-positive rate on the same images?

For RQ1, the targets are unseen F1 ≥ 0.80, false-positive rate ≤ 0.10 on each original-image test, and a seen-to-unseen F1 drop ≤ 0.10. For RQ2, we examine changes in both F1 and false positives, since a higher F1 alone does not establish robustness.

We refit both configurations on the entire available Tiny-GenImage mirror, retaining 34,998 images after exact deduplication and using six development generators. Saved scores allow each processing-induced error to be traced to the same test ID across three seeds. Paired intervals, decision-transition counts and threshold checks let us assess whether an F1 advantage accompanies acceptable false-positive rates. Previously inspected pilot content can be included in the new split, so the refit does not provide an independent confirmation cohort. The statistical methods are standard, the fitting budgets differ, and source differences remain uncontrolled. We treat the findings as exploratory.

2 Literature Review

Testing an unfamiliar generator, a new category from a familiar generator, and a compressed version of an existing input measures different capabilities. The studies below address these problems through surveys, transfer experiments and alternative detection features.

2.1 Reviews and evaluation scope

Foo et al. review generated media across several modalities [13], while Chen et al. cover diffusion-based generation and editing [14]. We limit our task to binary detection of ordinary still images. Video detection, source attribution and localization of partial edits need different labels and evaluation.

Zhang et al., Mahara and Rishe, and Xu et al. discuss spatial, frequency and multimodal evidence, including weaknesses on unfamiliar inputs [15–17]. Mahara and Rishe distinguish generator, category and scene shifts. Xu et al. separate transfer from robustness and explanation. We follow these distinctions by treating generator transfer and processing robustness as separate research questions.

Pei et al. compare facial benchmarks [18], and Heidari et al. review deep learning methods across media [19]. Facial and temporal clues need separate testing before they can be applied to object images. Shahzad et al. describe earlier detection methods [20]; Singh et al. discuss dataset bias and deployment costs [21]. Deng et al. examine detection alongside authentication, watermarking and disruption [22]. These reviews support reporting false positives and resource requirements together with detection scores.

2.2 Transfer beyond development data

Lamichhane tests pretrained vision transformers and examines the effects of pretraining and augmentation [23]. Saskoro et al. combine ResNet experts through a learned gate and evaluate additional generators [24]. Both studies motivate our use of pretrained representations and multiple development sources. Their internal scores, however, cannot predict performance on our held-out data.

Coccomini et al. compare convolutional and CLIP-feature classifiers on diffusion images paired with different photographic sources. Their transfer results depend on the direction of the source change [25]. Bayram reports strong internal CIFAKE performance but much weaker results on external Tiny-GenImage data, where predictions concentrate on the generated class [26]. That error pattern makes false-positive rates especially relevant. In both studies, external testing can change more than generator identity.

Rout et al. evaluate attention and convolutional features on external forgery datasets that differ in domain and forgery type [27]. Sanders et al. hold out categories while keeping Stable Diffusion XL as the generator [28]. These experiments test different forms of transfer. We explicitly hold out a generator, although differences in photographic sources and resolution still limit causal interpretation.

2.3 Additional features and uncertainty

Man and Cho combine CLIP content features with wavelets and attention [29]. Yu and Xu fuse image and texture information [30], while SCADET uses frequency attention and contrastive spectral analysis for artwork [31]. Their component tests suggest that texture can add information to semantic features. Findings from artwork still need testing on ordinary photographs, so texture fusion remains future work here.

Yumlembam et al. combine uncertainty measures with optimized rejection [32]. Rejecting uncertain inputs can remove errors while reducing the number of images that receive a decision; future tests of rejection should report that coverage. We use a fixed threshold and a validation-selected alternative, with no rejection. Both configurations are compared on shared IDs across three seeds, with compression and downsampling tested separately.

3 Materials and Methods

3.1 Data, allocation and provenance

We use the entire 35,000-row Tiny-GenImage mirror, a third-party subset related to GenImage [33, 34], pinned at revision `89c4fe9efd0e`. On 6 November 2026, source parquet shards were downloaded and checked against their published SHA256 digests. We used the embedded image bytes rather than the dataset viewer's image responses. The mirror's original training and validation partitions were pooled, then split with sampling seed 20261106. Global screening of exact files, decoded RGB pixels and common-encoding pixels removed two duplicate photographs. The retained cohort has 17,498 real and 17,500 generated images (Table 1). Perceptual near-duplicate screening and comparison with the official GenImage archives remain incomplete.

All 2,500 Midjourney images are reserved for the generated part of the unseen test; none enter training or validation. ADM, BigGAN, GLIDE, SD15, VQDM and Wukong each supply 2,500 generated images across training, validation and the seen test. SD14 has no rows in this mirror.

The seen test contains 500 images from each development generator. Real photographs are disjoint across the four partitions, as are partition IDs and exact file, decoded-pixel and common-encoding hashes. Within each test, both models, every seed and all processing conditions use the same IDs. This supports paired comparisons within a test; seen-to-unseen changes compare different cohorts.

The initial plan specified 20,000 images. We expanded the refit to the entire mirror and changed the data split, source access, development generators and fitted models. Previously inspected pilot content can enter the new partitions, so this is not an independent holdout of all previously studied source content. The earlier 1,920-image analysis remains in the repository as historical evidence and is not pooled with these results. Any difference from that pilot could reflect several protocol changes, not just sample size.

Before training or testing, all images are converted to RGB and re-encoded at native size as JPEG quality 75 with 4:2:0 chroma subsampling. We call this common quality-75 input the "original" condition. Prior compression, native size and content categories are audited but not matched. BigGAN images are 128×128; ADM, GLIDE and VQDM are 256×256; SD15 and Wukong are 512×512. Midjourney dimensions range from 512 to 1,024 pixels. Real-image widths range from 34 to 4,786 and heights from 38 to 4,419. A common final encoding leaves differences in source history and resolution that a detector could use as shortcuts [35].

Table 1 Study partitions before and after exact-file, decoded-RGB and common-encoding duplicate screening.

Partition	Nominal	Retained	Real	Generated
Training	20,000	19,998	9,998	10,000
Validation	4,000	4,000	2,000	2,000
Seen test	6,000	6,000	3,000	3,000
Unseen test	5,000	5,000	2,500	2,500
Total	35,000	34,998	17,498	17,500

Two duplicate photographs were removed before splitting. Midjourney occurs only in the unseen test; six other generators supply development and seen-test images. The cohort includes previously inspected pilot images.

3.2 Models and training

For ResNet-50, we start from torchvision ImageNet1K V1 weights, replace the final layer with one binary logit, and fine-tune all parameters [36, 37]. CLIP ViT-B/32 uses OpenAI weights. Its encoder stays frozen while a linear layer is trained on the 512-dimensional features [38, 39]. Inputs are resized directly to 224×224 by bicubic interpolation, without cropping, then normalized for each model. We use no data augmentation.

Both models use Adam and binary cross-entropy with logits. The positive-class weight is calculated from the training real/generated ratio, $9,998/10,000 = 0.9998$, without test labels. ResNet trains for one epoch with batch size 16 and learning rate 10^{-4}. The CLIP head trains for 40 epochs with batch size 32 and learning rate 10^{-3}. Seeds 42, 43 and 44 share the same manifest. We retain the checkpoint with the highest original-validation F1 at threshold 0.5, choosing the earliest epoch in a tie. CLIP selects epochs 38, 40 and 38; ResNet selects epoch 1 in every run. Because the fitting budgets differ, the results compare these configurations and cannot establish an architecture-only effect.

All runs used an AMD EPYC 9V74 CPU without a GPU. The encoders used CPU bfloat16 autocast, with float32 parameters and Adam state. CLIP-head training, cached features and saved scores used float32. The main pipeline used six CPU threads, CLIP feature preparation used two, and head fitting/scoring used one. Two ResNet seed-44 test groups used two threads. Separate stage metadata record these settings, and `checkpoints.json` identifies all six fitted checkpoints by SHA256. The supplied CUDA entry point was not used for these results.

3.3 Metrics and paired uncertainty

We treat generated images as positive. A false positive is a real image labelled generated. Using true-positive (TP), true-negative (TN), false-positive (FP) and false-negative (FN) counts, we calculate

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{Precision} = \frac{TP}{TP + FP}, \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2TP}{2TP + FP + FN}, \tag{2}$$

$$\text{FPR} = \frac{FP}{FP + TN}. \tag{3}$$

Undefined precision or F1 is set to zero. We also report area under the receiver operating characteristic curve (AUROC) and average precision (AP). Main results use threshold 0.5 and are reported as means and sample standard deviations (SDs) across the three fitted seeds.

We use a paired bootstrap stratified by label and source, with 1,000 resamples, to obtain percentile 95% confidence intervals. Each resample contains the same IDs for both models and all seeds; metrics are calculated per seed before averaging. The intervals describe image-sampling uncertainty conditional on the fitted checkpoints. SDs across the three seeds describe observed fitting variation, which is not resampled as a second level. Repeated evaluations of an image across seeds or conditions do not add independent images.

For the primary comparison on original unseen images, we use 9,999 two-sided paired permutations. Model outputs are exchanged within each image, jointly over seeds. If k absolute differences are at least as large as observed, $p = (1 + k)/10,000$. The smallest attainable value is 0.0001, the Monte Carlo resolution floor. Our practical comparison rule requires an absolute mean F1 difference ≥ 0.03, an interval that excludes zero in the same direction, and mean FPR ≤ 0.10 for the higher-F1 model in that condition. This rule assesses a difference between models separately from the combined absolute performance targets. Secondary comparisons across conditions are exploratory and unadjusted.

3.4 Processing and threshold sensitivity

We evaluate the fixed checkpoints on the common quality-75 baseline and on two separately processed versions: JPEG quality-65 recompression and downsampling to 112×112. Each then receives the usual 224×224 input resize. Compression and downsampling are applied independently. Paired changes are $\Delta F_1 = F_{1,\text{processed}} - F_{1,\text{original}}$ and $\Delta\text{FPR} = \text{FPR}_{\text{processed}} - \text{FPR}_{\text{original}}$.

To examine threshold sensitivity, we compare 0.5 with the threshold that maximizes F1 on the 4,000-image original validation split. We search separately for each model and seed from 0.05 to 0.95 in steps of 0.05, choosing the larger threshold in a tie. Test labels are not used for selection. This rule optimizes validation F1 without directly imposing an FPR constraint.

4 Results

The refit uses 34,998 images, of which 11,000 are distinct test images. No fitted seed meets all performance targets.

4.1 RQ1: Original images and false positives

ResNet's mean F1 falls from 0.910 on seen generators to 0.310 on Midjourney; CLIP's falls from 0.949 to 0.746 (Table 2). Calculated before rounding, the drops are 0.599 and 0.203, both above 0.10. Neither model reaches unseen F1 0.80. CLIP stays within the original-image FPR limit on both tests in every seed. ResNet's mean FPR is below 0.10, although seeds 43 and 44 exceed that limit on both original-image tests. The mean therefore hides failures in individual fitted runs.

Unseen precision and recall are 0.680 and 0.206 for ResNet, compared with 0.911 and 0.632 for CLIP. CLIP detects more Midjourney images and produces fewer false positives. ResNet's

seen-to-unseen AUROC falls from 0.971 to 0.652; CLIP's falls from 0.988 to 0.925. AP falls from 0.971 to 0.629 and from 0.986 to 0.908, respectively. CLIP's strong unseen ranking does not translate into equally strong F1 at threshold 0.5, so its operating point needs separate assessment.

Table 2 Original-image metrics at threshold 0.5. Values are means ± sample SD over seeds 42, 43 and 44. P is precision and R is recall.

Model	Test	Accuracy	P	R	F1
ResNet-50	Seen	0.909 ± 0.002	0.904 ± 0.024	0.916 ± 0.031	0.910 ± 0.004
CLIP linear	Seen	0.949 ± 0.001	0.942 ± 0.001	0.957 ± 0.002	0.949 ± 0.001
ResNet-50	Unseen	0.555 ± 0.025	0.680 ± 0.026	0.206 ± 0.084	0.310 ± 0.106
CLIP linear	Unseen	0.785 ± 0.002	0.911 ± 0.002	0.632 ± 0.007	0.746 ± 0.004

Model	Test	FPR	AUROC	F1 95% CI	FPR 95% CI
ResNet-50	Seen	0.098 ± 0.029	0.971 ± 0.002	[0.904, 0.916]	[0.089, 0.107]
CLIP linear	Seen	0.059 ± 0.001	0.988 ± 0.000	[0.944, 0.955]	[0.050, 0.068]
ResNet-50	Unseen	0.096 ± 0.038	0.652 ± 0.022	[0.296, 0.326]	[0.088, 0.105]
CLIP linear	Unseen	0.061 ± 0.002	0.925 ± 0.000	[0.732, 0.759]	[0.052, 0.071]

Seen test: 3,000 images per class. Unseen test: 2,500 per class. Confidence intervals resample images within label/source strata and condition on fitted checkpoints.

4.2 Paired model differences and uncertainty

On original unseen images, the CLIP-minus-ResNet F1 difference is 0.436, with a paired 95% interval of [0.415, 0.456] and permutation $p = 0.0001$ (Table 3). This exceeds 0.03, the interval excludes zero, and CLIP's mean unseen FPR is 0.061. The comparison therefore meets our practical rule. CLIP still fails the combined absolute targets: unseen F1 remains below 0.80 and the seen-to-unseen loss exceeds 0.10. Its advantage over ResNet does not make its absolute performance sufficient.

Absolute unseen F1 intervals are [0.296, 0.326] for ResNet and [0.732, 0.759] for CLIP. FPR intervals are [0.088, 0.105] and [0.052, 0.071], so ResNet's interval crosses the 0.10 limit. Unseen F1 SDs are 0.106 and 0.004, respectively. ResNet varies much more across the fitted seeds. Changing one real-image decision changes a run's unseen FPR by $1/2,500 = 0.0004$. These image-bootstrap intervals do not capture uncertainty from new training runs or data sources.

CLIP has higher mean F1 in all six test/condition comparisons. The gap is 0.040 on seen original images and 0.067 on resized unseen images. However, CLIP's FPR exceeds 0.10 after resizing, so neither resized comparison meets the practical rule.

Table 3 Paired F1 differences (CLIP minus ResNet-50), with image-bootstrap 95% intervals. The original-unseen comparison is primary; other p-values are exploratory and unadjusted.

Test	Condition	ΔF1	95% CI	p
Seen	Original	0.040	[0.033, 0.046]	0.0001
Seen	JPEG 65	0.021	[0.013, 0.030]	0.0001
Seen	Resize 112	0.082	[0.076, 0.089]	0.0001
Unseen	Original	0.436	[0.415, 0.456]	0.0001
Unseen	JPEG 65	0.398	[0.377, 0.419]	0.0001
Unseen	Resize 112	0.067	[0.059, 0.075]	0.0001

The Monte Carlo permutation resolution is 0.0001 (9,999 swaps plus the observed sample). An F1 advantage also needs an acceptable false-positive rate to meet the practical comparison rule.

4.3 RQ2: Recompression and downsampling

Every test image remains evaluable after processing. JPEG quality-65 recompression lowers unseen F1 to 0.289 for ResNet and 0.687 for CLIP. The paired changes, -0.022 and -0.059, have intervals excluding zero (Table 4). Unseen FPR also falls, to 0.062 and 0.028, with intervals for both reductions excluding zero. Recall declines to 0.183 for ResNet and 0.538 for CLIP. Recompression reduces false alarms at the cost of missed detections.

After downsampling, unseen ResNet F1 rises to 0.666 and CLIP F1 falls to 0.733. Their FPRs rise to 0.971 and 0.561. ResNet is close to the all-generated baseline, $F_1 = 2/3 \approx 0.667$, for this balanced test; its unseen accuracy is 0.506 and AUROC 0.561. CLIP retains more discrimination, with accuracy 0.671 and AUROC 0.800, but falsely flags over half of the photographs. Its paired F1 change is -0.013, with interval $[-0.027, 0.001]$. The FPR increase is 0.500, with interval [0.480, 0.519]. The relatively small F1 change therefore accompanies a large rise in false alarms.

On seen images, downsampling reduces F1 by 0.236 for ResNet and 0.193 for CLIP; FPR reaches 0.964 and 0.564. Both models fail the FPR requirement in every resized run. Figure 1 plots the unseen results with conditional image-bootstrap intervals. These tests cover only one recompression quality and one downsampling size.

Table 4 Paired processing minus original changes, averaged over seeds, with 95% image-bootstrap intervals. Compression and downsampling are applied independently to the common quality-75 baseline.

JPEG recompression, quality 65

Model	Test	ΔF_1 [95% CI]	ΔFPR [95% CI]
ResNet-50	Seen	-0.010 [-0.014, -0.007]	-0.034 [-0.039, -0.030]
CLIP linear	Seen	-0.029 [-0.035, -0.023]	-0.029 [-0.036, -0.023]
ResNet-50	Unseen	-0.022 [-0.026, -0.017]	-0.034 [-0.039, -0.030]
CLIP linear	Unseen	-0.059 [-0.070, -0.047]	-0.034 [-0.041, -0.027]

Downsampling to 112×112 pixels

Model	Test	ΔF_1 [95% CI]	ΔFPR [95% CI]
ResNet-50	Seen	-0.236 [-0.241, -0.230]	+0.866 [0.857, 0.876]
CLIP linear	Seen	-0.193 [-0.200, -0.185]	+0.505 [0.486, 0.522]
ResNet-50	Unseen	+0.355 [0.341, 0.372]	+0.874 [0.865, 0.884]
CLIP linear	Unseen	-0.013 [-0.027, 0.001]	+0.500 [0.480, 0.519]

Each comparison uses identical IDs within the 6,000-image seen or 5,000-image unseen test. An F1 rise does not establish robustness when false positives increase.

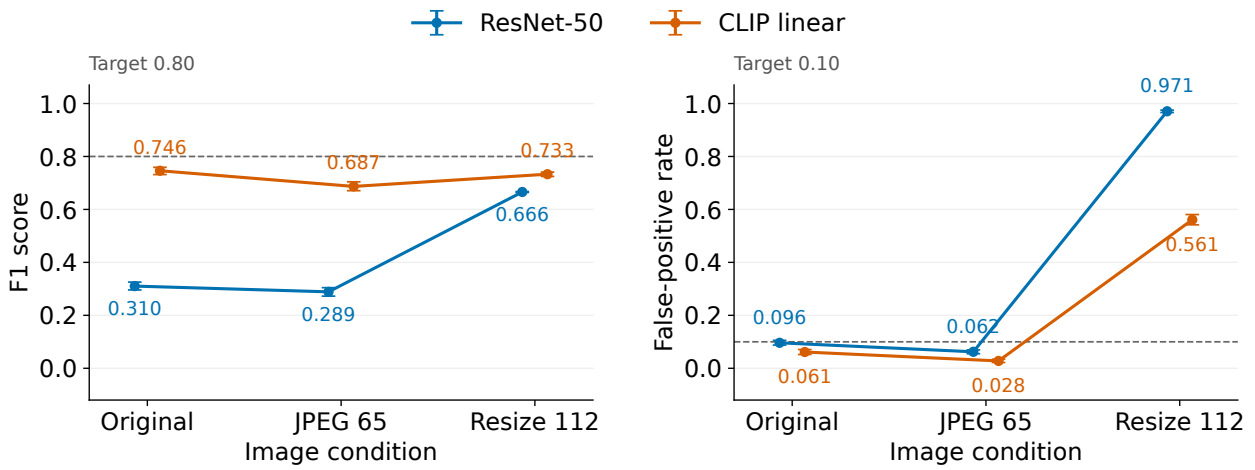


Figure 1 Full-mirror unseen-test means over three seeds. Error bars are stratified image-bootstrap 95% intervals conditional on fitted checkpoints. Each condition uses the same 2,500 real and 2,500 Midjourney images. Dashed lines mark minimum F1 0.80 and maximum FPR 0.10.

4.4 Threshold sensitivity and observed errors

Validation selects ResNet thresholds of 0.25, 0.55 and 0.55 for seeds 42, 43 and 44. CLIP selects 0.50, 0.55 and 0.50. ResNet's mean unseen F1 rises from 0.310 to 0.334, while FPR rises from 0.096 to 0.104. CLIP's mean unseen F1 falls from 0.746 to 0.737 and FPR falls from 0.061 to 0.058 (Table 5). Neither threshold rule reaches unseen F1 0.80. Selecting the best validation F1 leaves the transfer problem unresolved and does not ensure compliance with an FPR constraint.

Across seeds 42–44, ResNet misses 2,222, 1,822 and 1,908 of the 2,500 Midjourney images; CLIP misses 907, 917 and 939. ResNet produces 133, 279 and 310 false positives, compared with 158, 155 and 148 for CLIP. ResNet seed 42 thus has fewer false positives than CLIP seed 42, even though its mean FPR across the three seeds is higher. Figure 2 shows the seed-42 matrices, and the seed-specific exports retain all counts.

After unseen downsampling, 2,114–2,264 previously correct real-image decisions per seed become ResNet false positives, while 0–2 errors reverse. For CLIP, 1,239–1,276 correct real-image decisions become false positives and 11–13 reverse. Downsampling also corrects 1,800–2,172 prior Midjourney misses and CLIP corrects 690–709. These changes explain why increased detection can coincide with a large loss of specificity.

Table 5 Threshold sensitivity on original images: fixed threshold 0.5 versus the threshold maximizing F1 on the 4,000-image validation split. Values are means ± sample SD.

Model	Test	Rule	F1	FPR
ResNet-50	Seen	Fixed	0.910 ± 0.004	0.098 ± 0.029
ResNet-50	Seen	Validation	0.914 ± 0.001	0.106 ± 0.010
ResNet-50	Unseen	Fixed	0.310 ± 0.106	0.096 ± 0.038
ResNet-50	Unseen	Validation	0.334 ± 0.019	0.104 ± 0.007
CLIP linear	Seen	Fixed	0.949 ± 0.001	0.059 ± 0.001
CLIP linear	Seen	Validation	0.948 ± 0.001	0.057 ± 0.004
CLIP linear	Unseen	Fixed	0.746 ± 0.004	0.061 ± 0.002
CLIP linear	Unseen	Validation	0.737 ± 0.015	0.058 ± 0.006

Thresholds are selected separately per model/seed from 0.05 to 0.95 in steps of 0.05, with the larger threshold winning ties. Test labels never select the threshold.

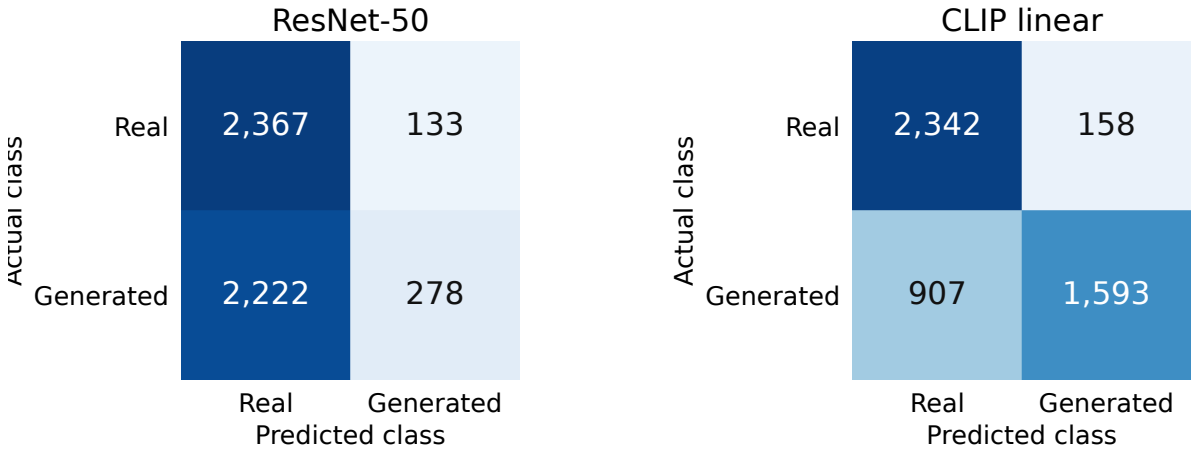


Figure 2 Original-unseen confusion matrices at threshold 0.5 for seed 42. Rows are actual classes and columns predicted classes. Each matrix contains 2,500 real and 2,500 Midjourney images.

4.5 Computation and reproducibility

The selected validation F1 scores are 0.907, 0.917 and 0.911 for ResNet, and 0.950, 0.950 and 0.951 for CLIP. Caching original CLIP features for all 34,998 images took 1,485.29 seconds. Preparation ran in parallel, training was resumed, and thread counts differed between stages, so this time does not support an architecture speed comparison. Recorded times for resumed epochs cover only their final resumed segments. They cannot be treated as complete epoch wall times or used to rank training speed.

The test export has 198,000 scores from evaluating 11,000 images with two models, three seeds and three conditions. Validation adds 24,000 scores for 4,000 images. These repeated evaluations do not increase the number of independent images. The archived score-based reproduction check regenerated 13 numerical exports byte for byte from the committed scores, manifest and protocol, without source images or fitted weights. The current five LaTeX tables and four figure files (two PNG/PDF pairs) were rendered from those numerical exports in two separate directories and matched byte for byte. Confusion-count metrics and seed means also agree with the archived exports. These checks verify reporting consistency without rerunning training.

5 Conclusion

For RQ1, CLIP detects more unseen generated images and has a lower mean false-positive rate than ResNet-50 after the refit on the entire available mirror. Mean unseen F1 is 0.746 for CLIP and 0.310 for ResNet, with FPR 0.061 and 0.096. CLIP's original-unseen advantage of 0.436 meets the practical paired comparison rule. Both models fall short of the absolute requirements: unseen F1 stays below 0.80, and seen-to-unseen F1 losses of 0.203 and 0.599 exceed 0.10. ResNet also violates the FPR limit in two of its three fitted seeds, despite a mean below the limit.

For RQ2, JPEG recompression reduces unseen F1 as well as false-positive rates. Downsampling has a much larger effect on false alarms: FPR reaches 0.561 for CLIP and 0.971 for ResNet. ResNet's resized F1 approaches the all-generated baseline. Although CLIP retains a higher resized F1, it falsely flags most photographs. A useful assessment therefore needs detection scores and false-positive rates for both original and processed inputs.

These results apply to the fitted configurations and this mirror. They do not establish readiness for practical verification or isolate a causal effect of generator identity. We use one third-party mirror and one held-out generator, with unequal fitting budgets and unmatched categories, resolution and compression history. Perceptual duplicates and possible pretraining exposure remain unresolved. Because previously inspected pilot content can enter the refit, we treat the findings as exploratory and do not claim independent confirmation. Bootstrap intervals condition on the fitted checkpoints,

and three seeds give limited evidence about variation under full retraining.

Further tests should audit the official archives, screen near-duplicates, match source characteristics, and use fresh external cohorts selected before their outcomes are inspected. Longer ResNet fitting with an explicit budget, more seeds, generator-wise holdouts and additional processing levels would help establish whether CLIP's advantage persists. Threshold selection should directly test an FPR-constrained operating point on development data. Any use in fact-checking would still need source and provenance checks, since a detector score alone cannot establish an image's origin.

6 Data Availability

Source images are available in `TheKernel01/Tiny-GenImage`, revision 89c4fe9efd0ebc7ce5c7641ef5 7d578ccd639c69, accessed 6 November 2026. The mirror declares Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0); the official-source audit is incomplete. The manifest, labels, source-shard records, exact-duplicate audit, test and validation scores, and numerical exports are public at https://github.com/C0nflu3nc3/RMT, under `experiments/full35k/`. The manifest SHA256 begins `1d8fb1ab8e174d31`, with the full digest in `sample_audit.json`. Source images, image caches, CLIP feature caches and large fitted weights are kept outside Git. Checkpoint digests identify the six selected fitted models. Committed scores support numerical reproduction; retraining needs the pinned source images and pretrained weights.

7 Code Availability

Code for selection, training, evaluation and reporting is public at https://github.com/C0nflu3nc3/RMT. The full-mirror evidence snapshot is `f9624d9cce60`, checked on 6 November 2026. Runs used Python 3.12.14, PyTorch 2.14.1+cpu, torchvision 0.29.1+cpu, openai-clip 1.0.1, NumPy 2.3.5, Pillow 13.0.0, scikit-learn 1.8.0 and SciPy 1.17.0. Training dependencies are pinned in `experiments/full35k/requirements.txt`, and analysis dependencies are in `requirements-analysis.txt`.

To reproduce metrics from the archived scores on Python 3.12, install the analysis requirements and run `python reproduce_tables.py` at the repository root. A new fit requires the training dependencies: run `collect.py`, then `train.py`, in `experiments/full35k/`. These steps download the pinned source and pretrained weights. The code uses an MIT licence; dataset and pretrained-model licences are separate. The earlier pilot and its historical analysis remain in `pilot/`, `expanded_test/`, `article/evaluation/` and Git history.

References

[1] B. Chandra, J. Dunietz, K. Roberts, Y. Lee, P. Fontana and G. Awad. Reducing risks posed by synthetic content: An overview of technical approaches to digital content transparency. NIST AI 100-4, 2024. doi:10.6028/NIST.AI.100-4.

[2] S. J. Nightingale and H. Farid. AI-synthesized faces are indistinguishable from real faces and more trustworthy. PNAS, 119(8):e2120481119, 2022. doi:10.1073/pnas.2120481119.

[3] A. Rössler et al. FaceForensics++: Learning to detect manipulated facial images. ICCV, pp. 1–11, 2019. doi:10.1109/ICCV.2019.00009.

[4] R. Corvi et al. On the detection of synthetic images generated by diffusion models. ICASSP, pp. 1–5, 2023. doi:10.1109/ICASSP49357.2023.10095167.

[5] J. Frank et al. Leveraging frequency analysis for deep fake image recognition. ICML, 119:3247–3258, 2020.

[6] R. Durall, M. Keuper and J. Keuper. Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions. CVPR, pp. 7887–7896, 2020. doi:10.1109/CVPR42600.2020.00791.

[7] C. Tan et al. Learning on gradients: Generalized artifacts representation for GAN-generated images detection. CVPR, pp. 12105–12114, 2023. doi:10.1109/CVPR52729.2023.01165.

[8] C. Tan et al. Rethinking the up-sampling operations in CNN-based generative network for generalizable deepfake detection. CVPR, pp. 28130–28139, 2024. doi:10.1109/CVPR52733.2024.02657.

[9] Z. Wang et al. DIRE for diffusion-generated image detection. ICCV, pp. 22388–22398, 2023. doi:10.1109/ICCV51070.2023.02051.

[10] J. Ricker, D. Lukovnikov and A. Fischer. AEROBLADE: Training-free detection of latent diffusion images using autoencoder reconstruction error. CVPR, pp. 9130–9140, 2024. doi:10.1109/CVPR52733.2024.00872.

[11] H. Liu et al. Forgery-aware adaptive transformer for generalizable synthetic image detection. CVPR, pp. 10770–10780, 2024. doi:10.1109/CVPR52733.2024.01024.

[12] D. Cozzolino et al. Raising the bar of AI-generated image detection with CLIP. CVPR Workshops, pp. 4356–4366, 2024. doi:10.1109/CVPRW63382.2024.00439.

[13] L. G. Foo, H. Rahmani and J. Liu. AI-generated content (AIGC) for various data modalities: A survey. ACM Computing Surveys, 57(9):243, 2025. doi:10.1145/3728633.

[14] H. Chen et al. Comprehensive exploration of diffusion models in image generation: A survey. Artificial Intelligence Review, 58(4):99, 2025. doi:10.1007/s10462-025-11110-3.

[15] Y. Zhang et al. Unmasking AI-created visual content: A review of generated images and deepfake detection technologies. Journal of King Saud University Computer and Information Sciences, 37(6):148, 2025. doi:10.1007/s44443-025-00154-8.

[16] A. Mahara and N. Rishe. Methods and trends in detecting AI-generated images: A comprehensive review. Computer Science Review, 60:100908, 2026. doi:10.1016/j.cosrev.2026.100908.

[17] Q. Xu et al. Advancements in AI-generated content forensics: A systematic literature review. ACM Computing Surveys, 58(3):60, 2025. doi:10.1145/3760526.

[18] G. Pei et al. Deepfake generation and detection: A benchmark and survey. ACM Computing Surveys, 58(11):273, 2026. doi:10.1145/3801962.

[19] A. Heidari et al. Deepfake detection using deep learning methods: A systematic and comprehensive review. WIREs Data Mining and Knowledge Discovery, 14(2):e1520, 2024. doi:10.1002/widm.1520.

[20] H. F. Shahzad et al. A review of image processing techniques for deepfakes. Sensors, 22(12):4556, 2022. doi:10.3390/s22124556.

[21] L. H. Singh, P. Charanarur and N. K. Chaudhary. Advancements in detecting deepfakes: AI algorithms and future prospects - a review. Discover Internet of Things, 5(1):53, 2025. doi:10.1007/s43926-025-00154-0.

[22] J. Deng et al. A survey of defenses against AI-generated visual media: Detection, disruption, and authentication. ACM Computing Surveys, 58(5):123, 2026. doi:10.1145/3770916.

[23] D. Lamichhane. Advanced detection of AI-generated images through vision transformers. IEEE Access, 13:3644–3652, 2025. doi:10.1109/ACCESS.2024.3522759.

[24] R. A. F. Saskoro, N. Yudistira and T. N. Fatyanosa. Detection of AI-generated images from various generators using gated expert convolutional neural network. IEEE Access, 12:147772–147783, 2024. doi:10.1109/ACCESS.2024.3466614.

[25] D. A. Coccomini et al. Detecting images generated by diffusers. PeerJ Computer Science, 10:e2127, 2024. doi:10.7717/peerj-cs.2127.

[26] F. Bayram. Color space analysis for AI-generated image detection on CIFAKE: CNN models and statistical evaluation. IEEE Access, 14:30747–30757, 2026. doi:10.1109/ACCESS.2026.3665315.

[27] J. Rout et al. Stacked multi-fusion CNN: An adaptive attention model for privacy preserving deepfake forensics. Scientific Reports, 16:24653, 2026. doi:10.1038/s41598-026-55325-y.

[28] B. Sanders, D. Morrison and D. Harris-Birtill. Exploring techniques to distinguish between real images and those generated using Stable Diffusion XL. PLOS ONE, 21(1):e0339917, 2026. doi:10.1371/journal.pone.0339917.

[29] Q. Man and Y.-I. Cho. Transformer based on multi-domain feature fusion for AI-generated image detection. Electronics, 15(3):716, 2026. doi:10.3390/electronics15030716.

[30] H. Yu and B. Xu. Multi-modal texture fusion network for detecting AI-generated images. Frontiers in Artificial Intelligence, 8:1663292, 2025. doi:10.3389/frai.2025.1663292.

[31] X. Zhang, Z. Yu and J. Zhao. SCADET: A detection framework for AI-generated artwork integrating dynamic frequency attention and contrastive spectral analysis. PLOS ONE, 20(11):e0336328, 2025. doi:10.1371/journal.pone.0336328.

[32] R. Yumlembam et al. Detection of AI generated images using combined uncertainty measures and particle swarm optimised rejection mechanism. Scientific Reports, 15:44021, 2025. doi:10.1038/s41598-025-28572-8.

[33] M. Zhu et al. GenImage: A million-scale benchmark for detecting AI-generated image. NeurIPS, 36:77771–77782, 2023. doi:10.52202/075280-3398.

[34] TheKernel01. Tiny-GenImage dataset. Revision 89c4fe9efd0e; accessed 6 November 2026. Mirror-declared licence CC BY-NC-SA 4.0. https://huggingface.co/datasets/TheKernel01/Tiny-GenImage.

[35] P. Grommelt et al. Fake or JPEG? Revealing common biases in generated image detection datasets. ECCV 2024 Workshops, LNCS 15644, pp. 80–95, 2025. doi:10.1007/978-3-031-92089-9_6.

[36] K. He et al. Deep residual learning for image recognition. CVPR, pp. 770–778, 2016. doi:10.1109/CVPR.2016.90.

[37] S.-Y. Wang et al. CNN-generated images are surprisingly easy to spot... for now. CVPR, pp. 8692–8701, 2020. doi:10.1109/CVPR42600.2020.00872.

[38] A. Radford et al. Learning transferable visual models from natural language supervision. ICML, 139:8748–8763, 2021.

[39] U. Ojha, Y. Li and Y. J. Lee. Towards universal fake image detectors that generalize across generative models. CVPR, pp. 24480–24489, 2023. doi:10.1109/CVPR52729.2023.02345.